

End-to-end fixed- L Gram:

the first block SATZ and the open tail

PRIME.RDAGGER.ENDTOEND_FIXEDL.01 — round 478

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Abstract

At $L = 0.3$ one has $P \equiv 0$ and $t_0 = 22$. The even Fourier Gram of Q_W on $\text{span}\{\cos(n\pi x/L) : 0 \leq n < N\}$ is interval-certified positive definite for $N = 2, 3, 4, 5$, with every constant sealed. The infinite tail Schur does not close: the constant-row couplings decay like $1/n$ and the Hilbert–Schmidt correction exceeds the Higham floor of the 3×3 . This is a SATZ on a finite-dimensional subspace, not $\lambda_*(0.3) \geq 0$ on $L^2[-L, L]$. At $L = 0.8$ the even Fourier 3×3 with exact rank-3 P is indefinite; the best certified substructure remains the r477 2×2 of (constant, first Dirichlet). The oscillatory multiplier $\sigma_P(t) = \sum 2\Lambda(n)n^{-1/2} \cos(t \log n)$ is almost periodic: high-band gain versus $2S_{\text{eff}}$ is at most 1.010 at $L = 0.8$ and does not unlock $L > 1.6$. Verdict **PARTIAL(0.3-even- 5×5 -PD|tail-Schur)**. No RH claim.

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Claim boundary. Nothing here is $\lambda_*(L) \geq 0$ on all of $L^2[-L, L]$. Finite even Fourier blocks are PD. The tail Schur is open (explicit HS obstruction). No silent class cut: the Paley–Wiener ideal $\{\hat{h} = 0 \text{ on } |t| < t_0\}$ meets compact support only at 0; aliasing of interval Fourier modes is measured, not wished away. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

1 Convention

After r461/r471, $\text{supp } h \subset [-L, L]$, $g = h * \tilde{h}$, $\text{supp } g \subset [-2L, 2L]$, $Q_W = A - P + \Pi$ with $P(g) = \sum_{n \leq e^{2L}} 2\Lambda(n)n^{-1/2}g(\log n)$ (scramble-sensitive) and Π of rank at most two. Plancherel as in r477: $A(g) = (2\pi)^{-1} \int \sigma_A(t)|\hat{h}(t)|^2 dt$, $\sigma_A(t) = \text{Re } \psi(\frac{1}{4} + it/2) - \log \pi$. Prime range $n \leq e^{2L}$, not e^L .

2 Even–odd split and the t_0 -lemma

Lemma 1 (Even–odd at $P \equiv 0$). *At $L = 0.3$ there is no prime-power node with $\log n \leq 0.6$, so $P \equiv 0$. A is a Fourier multiplier, hence $A(\text{even}, \text{odd}) = 0$. The bilinear Π -density of an odd g vanishes. Therefore $Q_W = Q_{\text{even}} \oplus Q_{\text{odd}}$ on $L^2[-L, L]$.*

Lemma 2 (t_0 and c_* , r477). $t_0(0.3) = 22$, $\sigma_A(22)_{\text{lo}} = 1.2517 \geq 1.2281 = 4 \sinh(0.3) + c_*$ with $c_* = 0.01$, $S_{\text{eff}} = 0$. *On the ideal class $\{\hat{h} = 0 \text{ on } |t| < t_0\}$ one has $Q_W \geq c_* \|h\|_2^2$. That class intersects compact support only at $h = 0$.*

3 The block SATZ

Write $u_0 = 1/\sqrt{2L}$ and $u_n = \cos(n\pi x/L)/\sqrt{L}$ ($n \geq 1$), an ONB of even $L^2[-L, L]$. The autocorrelation of u_n ($n \geq 1$) is $g_n(t) = (L - t/2) \cos(\omega t) - \sin(\omega t)/(2\omega)$ with $\omega = n\pi/L$ (minus sine; the r476 first Dirichlet uses a plus). The 0– n correlation is the single (not doubled) kernel $g_{0n}(t) = (-1)^{n+1} \sin(\omega t)/\omega$. $A + \Pi$ is the Bose series of r476/r477 with algebraic ψ -tail and cubic pad, $K = 200$.

Theorem 1 (Even Fourier blocks at $L = 0.3$). *Let $L = 0.3$ and $G^{(N)}$ be the $N \times N$ Gram of Q_W in the basis $\{u_0, \dots, u_{N-1}\}$.*

1. $N = 2$: the interval characteristic polynomial gives $\lambda_{\min}(G^{(2)}) \in [1.731281 \cdot 10^{-2}, 1.825107 \cdot 10^{-2}]$. Higham (raw) $\mu \geq 8.301584 \cdot 10^{-3}$.
2. $N = 3$: Higham (raw) $\mu \geq 2.457362 \cdot 10^{-3}$; midpoint $\lambda_{\min} \approx 1.239183 \cdot 10^{-2}$.
3. $N = 4, 5$: r465 inverse-Cholesky congruence certifies $G^{(N)} \succ 0$; midpoint $\lambda_{\min} \approx 1.070798 \cdot 10^{-2}$ and $1.027458 \cdot 10^{-2}$.

The constant-mode Rayleigh is $r(u_0) \in [1.58360545 \cdot 10^{-1}, 1.58360546 \cdot 10^{-1}]$. The first odd mode $\sin(\pi x/L)$ has $r \geq 2.996973 \cdot 10^{-1}$.

Remark 1. The 3×3 Higham floor is conservative (radius row-sum). Midpoint eigenvalues decrease toward the r476 first-Dirichlet Rayleigh $r \in [1.163 \cdot 10^{-2}, 1.180 \cdot 10^{-2}]$, which lies in the closed even span but not in any finite Fourier block.

4 Aliasing and the open tail

Interval cosine u_n is not bandlimited. Low-frequency mass $E_n = (2\pi\|u_n\|_2^2)^{-1} \int_{|t| < 22} |\hat{u}_n|^2$: $E_0 \approx 0.950$, $E_1 \approx 0.880$, $E_2 \approx 0.493$, $E_3 \approx 0.0119$. The low block $N \leq 3$ owns the band; $n \geq 3$ is already high.

Lemma 3 (Tail Schur obstruction). *The Hilbert–Schmidt mass of the couplings from the 3-block to modes $n = 3, \dots, 8$ satisfies $\|G_{\ell h}\|_F \geq 0.337$ (outward hulls). The diagonal lower bound $r_3 \geq 1.678$ gives a Schur correction $\geq 6.783 \cdot 10^{-2}$, which exceeds the Higham floor $2.457 \cdot 10^{-3}$ of $G^{(3)}$. Gershgorin on row 0 fails (harmonic $1/n$ couplings). The infinite Gram is therefore not certified PD by this Schur route.*

Π is rank one on the even sector ($\Pi(h) = 2|\langle h, \cosh(x/2) \rangle|^2$) and is computed exactly on every finite block; it is not the obstruction. The obstruction is $A_{\ell h}$ (sinc overlap), not a crude $4 \sinh L$ majorant.

5 $L = 0.8$: cost and best substructure

$t_0(0.8) = 4300$, Paley–Wiener count $N_{\text{even}} \approx 1095$, both parities $N \approx 2190$. Interval Cholesky of a 2190×2190 Bose Gram is $\sim 3.50 \cdot 10^9$ float flops plus $\sim 2 \cdot 10^6$ interval Bose entries: not a this-round computation.

Theorem 2 (Best $L = 0.8$ substructure). *The even Fourier 3×3 at $L = 0.8$ with P evaluated exactly at the three nodes $n = 2, 3, 4$ (linear functionals $g(\log n)$, not majorized) is indefinite (midpoint $\lambda_{\min} \approx -4.72 \cdot 10^{-3}$). The r477 Gram of $\text{span}\{1, \cos(\pi x/(2L))\}$ remains positive definite: $\text{Schur}_{10} = 7.413828 \cdot 10^{-2}$. The constant-mode Rayleigh is $r(1) = 8.027405 \cdot 10^{-2}$. Not $\lambda_*(0.8) \geq 0$.*

6 Oscillatory P

As a Fourier multiplier, $P = (2\pi)^{-1} \int \sigma_P(t) |\hat{h}(t)|^2 dt$ with

$$\sigma_P(t) = \sum_{n \leq e^{2L}} 2\Lambda(n) n^{-1/2} \cos(t \log n).$$

The symbol is a finite cosine sum, hence almost periodic. Its essential supremum on $[t_0, \infty)$ equals (up to an arbitrarily small gap) its essential supremum on $\mathbb{R}$: there is no $1/t_0$ decay of $\|\sigma_P\|_\infty$.

Lemma 4 (Measured gain, scramble-sensitive). *Triangle:* $|\sigma_P| \leq 2S_{\text{eff}}$. *Sampling* $[t_0, t_0 + 200]$ gives an upper bound on the gain $2S_{\text{eff}}/\|\sigma_P\|_\infty$: $L = 0.8$ (3 nodes) gain ≤ 1.0034 ; $L = 1.2$ (8 nodes) gain ≤ 1.133 ; $L = 1.8$ (18 nodes) gain ≤ 1.836 . A van der Corput saving for a single highly oscillatory g is a different pairing and is not an operator-norm bound.

7 Updated forecast

The r477 crude law $t_0 \sim 2\pi \exp(2S_{\text{eff}} + 4 \sinh L)$ is unchanged in order. Replacing $2S_{\text{eff}}$ by the measured $\|\sigma_P\|_\infty$ multiplies the exponent by $1/\text{gain} \approx 1$ at $L = 0.8$ and ≈ 0.54 at $L = 1.8$; $t_0(1.8)$ drops from $\sim 10^{13}$ to still $\sim 10^{10}$. L_{16} remains $\sim 10^{39}$. $L > 1.6$ is not unlocked.

8 Sealed balance

Probe `endtoend_fixedl_probe.py`, SPEC_SHA 138cbf9410486418. Smoke 17/17 in 0.31 s; full 27/27 in 3.9 s. Two smoke runs pin-identical (elapsed excluded).

object	type	sealed constant
$t_0(0.3)$	r477 lemma	22
$G^{(2)} \lambda_{\min}$	interval charpoly	$[1.731 \cdot 10^{-2}, 1.826 \cdot 10^{-2}]$
$G^{(2)} \text{Higham } \mu$	raw	$\geq 8.302 \cdot 10^{-3}$
$G^{(3)} \text{Higham } \mu$	raw	$\geq 2.457 \cdot 10^{-3}$
$G^{(4)}, G^{(5)}$	congruence PD	midpoint $1.071, 1.027 \cdot 10^{-2}$
tail HS correction	obstruction	$\geq 6.783 \cdot 10^{-2}$
$L = 0.8$ Fourier 3×3	indefinite	hint $-4.72 \cdot 10^{-3}$
$L = 0.8$ Dir. 2×2	r477 Schur _{lo}	$7.414 \cdot 10^{-2}$
K_P gain $L = 0.8$	$\leq$ triangle	≤ 1.010
$N(0.8)$ both parities	cost	2190, $\sim 3.50 \cdot 10^9$ flops

9 Kill table

gate	status	note
A-constants sealed	PASS	every number above is a pin
aliasing honest	PASS	E_n measured; no class cut
tail Schur	KILL	HS > Higham3; no λ_*
B full 2190 Chol	KILL (cost)	forecast recorded
B Fourier 3×3	KILL (indefinite)	exact P , not majorized
C $1/t_0$ gain	KILL	almost-periodic multiplier
scramble	PASS	literal $\log n$ in P and σ_P
determinism	PASS	two-run pins identical
anti-list / no RH	PASS	fixed L , e^{2L} , no zeros

10 What this does not do

It does not prove $\lambda_*(0.3) \geq 0$ or $\lambda_*(0.8) \geq 0$. It does not close the Paley–Wiener remainder as a complete-space argument. It does not unlock $L > 1.6$. No cofinal quantifier. No RH claim.